

Multilingual Signboard Translator Using OCR

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Abstract—In today's diverse linguistic landscape, language barriers can hinder communication, especially in multilingual countries like India, where travelers often encounter signboards in unfamiliar regional languages. This paper introduces the development of a Multilingual Signboard Translator Using OCR. This web-based application leverages Optical Character Recognition (OCR), Speech Recognition, Machine Translation, and Text-to-Speech (TTS) technologies to bridge language gaps in real-time. The system allows users to either upload images of signboards or speak live through a microphone to input text. The image-based input utilizes Tesseract OCR to detect and extract multilingual text, while the speech input uses Google Speech Recognition for real-time transcription. Extracted or transcribed text is then translated to the user-selected target language using Google Translate API, and the translated output is synthesized into speech using Google Text-to-Speech (gTTS), enabling both textual and auditory delivery. The application includes an image clarity validation step to prevent inaccurate OCR results. This system has practical implications for tourists, non-native residents, and field personnel, facilitating an accessible and seamless understanding of foreign-language signboards in real-world scenarios.

Keywords—OCR, Text-to-Speech (TTS), Signboard, Speech-to-Text (STT), Text Detection and Recognition, Multilingual Translation.

I. INTRODUCTION

India, being a linguistically diverse country, is home to 22 officially recognized languages and hundreds of regional dialects. While this diversity is culturally enriching, it poses significant communication challenges for travelers, migrants, and non-native speakers, who often encounter public information, such as signboards, in unfamiliar languages. This linguistic gap can hinder navigation, reduce accessibility, and create confusion in public spaces such as railway stations, bus terminals, roads, and tourist destinations.

With advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), enabling real-time translation and text recognition technologies have become more accurate and accessible. The combination of these technologies has opened up possibilities for intelligent systems that can assist users in understanding foreign-language content effortlessly. This project leverages these capabilities to develop a Multilingual Signboard Translator. This user-friendly web application translates text from images or live speech into a preferred language and delivers the output in both textual and auditory formats.

The system accepts two types of inputs: (1) an image of a signboard, or (2) live speech, ensuring that only one mode is active at a time to prevent conflicts. For image input, Tesseract OCR is used to detect and extract text in multiple Indian languages, including English, Hindi, Telugu, Tamil, Bengali, Kannada, Malayalam, and Marathi. For voice input, Google Speech Recognition transcribes the spoken content into text. This extracted or transcribed text is then translated using the Google Translate API into the user-selected language. The final translated result is converted into speech utilizing Google Text-to-Speech (gTTS), making it accessible to users with visual impairments or reading difficulties.

In addition to supporting multiple Indian languages, the system includes a clarity check for images to ensure reliable OCR output. If the image is too blurry, the system notifies the user, thus enhancing the overall accuracy and reliability of the application.

This tool is especially valuable in enhancing accessibility and inclusiveness for individuals with language or reading difficulties, while also offering practical use cases for travelers, field workers, and mobile application developers focused on smart city initiatives and intelligent transportation systems.

II. PROBLEM IDENTIFICATION & OBJECTIVES

Language barriers continue to be a major obstacle in accessing public information, especially in a linguistically diverse nation like India. Public signboards are often written in regional languages, making it difficult for non-native speakers, tourists, and inter-state travelers to understand crucial information. This becomes even more challenging in rural or semi-urban areas where English translations are not always provided. Current translation tools mostly focus on typing text manually or translating only from one language to another, and do not offer seamless OCR integration with support for multiple Indian scripts. Furthermore, existing solutions often lack audio output, reducing their accessibility for visually impaired users or those with low literacy levels.

To address these challenges, this project proposes the development of a web-based solution that combines multilingual OCR, machine translation, and speech synthesis into a single, accessible platform. The primary objective of this project is to develop a Multilingual Signboard Translator that can extract text from either an uploaded image of signboards or live spoken input and translate it into a user-selected language, followed by speech synthesis of the translated content. The system aims to support multiple Indian languages for both input and output, providing a seamless and accessible communication experience for a diverse range of users. Additional objectives include ensuring image clarity validation before performing OCR, allowing only one input mode at a time (image or speech), and delivering both textual and audio-based results. By combining OCR, machine translation, and TTS technologies into a single web application, the project seeks to bridge language gaps in real-time and promote inclusivity across language barriers.

III. SYSTEM METHODOLOGY

The proposed system follows a modular and systematic approach to provide a seamless multilingual signboard translation experience. It integrates state-of-the-art technologies such as Optical Character Recognition (OCR), Speech Recognition, Machine Translation, and Text-to-Speech synthesis to cater to both image and live audio inputs. The design ensures the system is user-friendly, flexible, and efficient, delivering translations in multiple Indian languages and converting them into speech.

IV. ARCHITECTURE

The first phase of the system involves image upload and processing. Users can upload an image containing text, such as a signboard, through the web interface. Upon receiving the image, the system performs an image clarity check using a sharpness measure (Laplacian Variance). If the image is too unclear for accurate text extraction, the system notifies the user with an appropriate error message. If the image passes the clarity check, the system extracts the text using Tesseract OCR, which is capable of recognizing text in multiple languages such as English, Hindi, Tamil, Bengali, Telugu, Kannada, Malayalam, and Marathi.

Alternatively, the system offers the option for live audio input. Users can communicate directly with the system using a microphone. The speech is processed using the Google

Speech Recognition API to convert the spoken words into text. In cases where the speech is not recognized or there is significant background noise, the system alerts the user with an error message such as "Could not understand audio." This flexibility allows the system to handle both image-based and audio-based inputs efficiently.

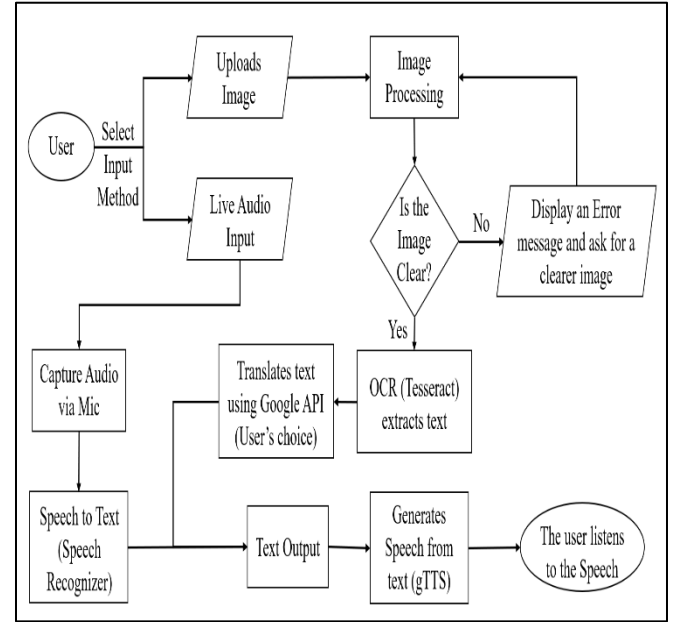


Fig. 1: Architecture

Once the text is extracted from the image or live audio, the system proceeds to translate it into the desired target language. The Google Translate API is employed to perform the translation, supporting a wide range of languages. The user can select their preferred language from the available options, ensuring personalized translations. Once the translation is complete, the system utilizes the Google Text-to-Speech (gTTS) API to convert the text translated into audio. This synthesized speech is saved as an audio file, which can then be downloaded or played back directly through the user interface.

The system provides real-time feedback, updating the user with the extracted text, translated text, and an audio file of the translated speech. If there are issues such as unclear images or unrecognized speech, appropriate error messages are displayed to guide the user. This ensures that the system is not only functional but also responsive to user needs.

The system offers a robust solution for multilingual signboard translation, leveraging advanced technologies to deliver accurate translations and audio outputs. Its modular approach guarantees flexibility and scalability, making it suitable for a wide variety of use cases in multilingual environments. By supporting both image and live audio inputs, the system ensures accessibility and usability for users in diverse linguistic contexts.

V. IMPLEMENTATION

A. Import Required Libraries

The first step in implementing the system is to set up the necessary libraries. The system requires the following libraries:

1. Flask: Used to create the web application.
2. cv2 (OpenCV): Used for image processing (checking clarity, reading images).
3. pytesseract: Utilizes OCR (Optical Character Recognition) to extract text from images.
4. Speech Recognition: for live audio input processing.
5. googletrans: Translates extracted text into the selected language.
6. gTTS (Google Text-to-Speech): Converts translated text into speech.
7. os: Handles file paths and directory creation.
8. time: Helps generate unique filenames for audio files.

B. Flask Application Setup

The Flask app is launched by utilizing the Flask class, which establishes a minimal web server to manage incoming requests. The app's routing system listens for GET and POST requests on the root endpoint (/). The POST method is used to process the user's input, which can either be an image or live audio. The application provides two main functionalities: extracting text from uploaded images and processing live audio input to extract speech.

Upon receiving a request, the system processes the uploaded content (either an image or audio), extracts the relevant information, translates it into the desired language, and provides the output both as translated text and an audio file. The Flask routes also handle user interaction, such as receiving the target language for translation and ensuring that the input type is validated before processing.

C. Image Clarity Check

Before attempting Optical Character Recognition (OCR), the system performs an image clarity check to ensure that the uploaded image contains sufficient visual quality for accurate text extraction. This check is done using the Laplacian Variance method in OpenCV, which calculates the image's sharpness by evaluating its variance. A high variance indicates that the image is sharp and clear, while a low variance suggests that the image is blurry or unclear. If the image's clarity is below the defined threshold, the system responds with a message notifying the user that the image is not clear enough for OCR.

Verifying the clarity of the image is crucial since the accuracy of OCR is largely influenced by the quality of the source image. Unclear or distorted visuals may result in inaccurate or partial text retrieval. By filtering out such images early in the process, the system ensures that only clear images are processed, thereby improving the reliability and accuracy of the text extraction step.

D. Text Extraction Using OCR

Once the image passes the clarity check, the system uses Pytesseract to perform Optical Character Recognition (OCR) and extract text from the image. Pytesseract is a powerful OCR tool that supports various languages, and it is used to recognize characters from the uploaded image. The application accommodates various Indian and international languages, such as English, Hindi, Tamil, Bengali, Telugu, Kannada, Malayalam, and Marathi, among others.

The system extracts text for each language supported by Tesseract. The information is kept in a dictionary where the key signifies the language code, and the value holds the identified text. This multi-language support ensures that the system can handle a wide range of signboards in different languages. Once the text is extracted, the system checks whether any recognizable text was found, and if not, it returns an appropriate message indicating that no text was detected.

E. Live Audio Input Processing

For live audio input, the system utilizes the Speech Recognition library, which allows real-time speech-to-text conversion. The system listens to the microphone input when the user presses the "Speak Now" button on the web interface. The audio is processed using Google's speech recognition service, which is capable of recognizing a wide range of languages and accents. The system adjusts for ambient noise to ensure that the audio input is captured clearly and accurately.

If the system successfully recognizes speech, the generated text is provided for translation. If the speech is not intelligible, or if there is a problem with the speech recognition service, the system responds with an error message such as "Could not understand audio" or "Speech recognition service unavailable." This live audio input feature provides users with an alternative to image-based input, allowing for greater flexibility in how text is provided for translation.

F. Text Translation

Once the text is extracted from the image or live audio input, the next step is translating it into the user's selected language. The system utilizes the Google Translate API for its translation processes. The user selects the desired language from a dropdown list on the web interface, and the system sends a request to the API to perform the translation. Google Translate supports a large number of languages, making it an ideal tool for this multilingual system.

The user is then provided with the translated text. In case of errors or if the translation is unavailable, the system ensures that a default message is shown. If the translation is successful, the system delivers the translated text along with the original extracted text in the response. This translation functionality supports a wide range of languages, including those commonly used for signboards and public signage in various regions.

G. Text-to-Speech Conversion

Once the text has been translated, the system utilizes the gTTS (Google Text-to-Speech) API to transform the translated content into spoken words. The gTTS library produces an MP3 audio file based on the provided text and is compatible with multiple languages. Once the text is converted into speech, the audio file is saved in the static/audio folder, where it can be accessed by the user.

The audio file is named dynamically using a timestamp to ensure that each generated speech file has a unique name. Once the speech is synthesized, the system generates a URL for the audio file, which is included in the JSON response. This enables the user to either download the audio or stream

it directly from the browser. The text-to-speech functionality enhances the user experience by providing an auditory translation of the signboard text, making the system more accessible.

H. Web Interface

The web interface is designed to be simple and user-friendly, allowing users to interact with the system intuitively. The main page (index.html) contains a form with options for the user to upload an image or speak live. The user can also select the target language for translation from a dropdown menu. The form handles both image uploads and live audio input, and the system ensures that only one input type is processed at a time.

The interface also displays the results after processing, showing the extracted text, translated text, and an audio player for the translated speech. The audio player allows users to listen to the translated text, and the user can also download the generated speech file for offline use. The clean and simple design ensures that users can easily navigate the system and obtain translations and speech outputs quickly.

I. Handling Requests and Responses

When the user submits the form, the system checks the type of input received. If an image is uploaded, the system first verifies its clarity before extracting text using OCR. If live audio is selected, the system uses the Speech Recognition library to capture and transcribe the speech. The system also validates that one input method is used at a time and ensures that all required information, such as the target language, is provided.

Once the input is processed, the system converts the extracted text into the chosen language with the help of Google Translate and creates a related speech file using gTTS. The results are returned to the user in the form of a JSON response containing the detected language, extracted text, translated text, and a link to the generated audio. The system also handles errors gracefully, providing appropriate messages if no text is detected or if an error occurs during any part of the process.

J. Audio Playback

The audio playback functionality allows users to listen to the translated text in the target language. Once the text is converted into speech, the system provides an audio player on the web interface to play the generated MP3 file. This feature enhances the usability of the system, as users can directly hear the translated text rather than only viewing it.

In addition to playing the audio directly on the web page, users can also download the generated speech file. This allows users to listen to the translated text offline, which can be useful in scenarios where internet connectivity is limited or unavailable. The audio file is made available via a separate route in the Flask application, ensuring smooth access and download for users.

VI. RESULTS

The system seamlessly integrates multiple components, including Optical Character Recognition (OCR) for text extraction from images, speech recognition for live audio

input, and translation capabilities for various Indian languages. Additionally, the system supports text-to-speech output in the selected language, enhancing accessibility and user experience. The following images demonstrate the system's core functionality and its ability to process both image and voice inputs, providing accurate translations in real-time across a variety of languages.

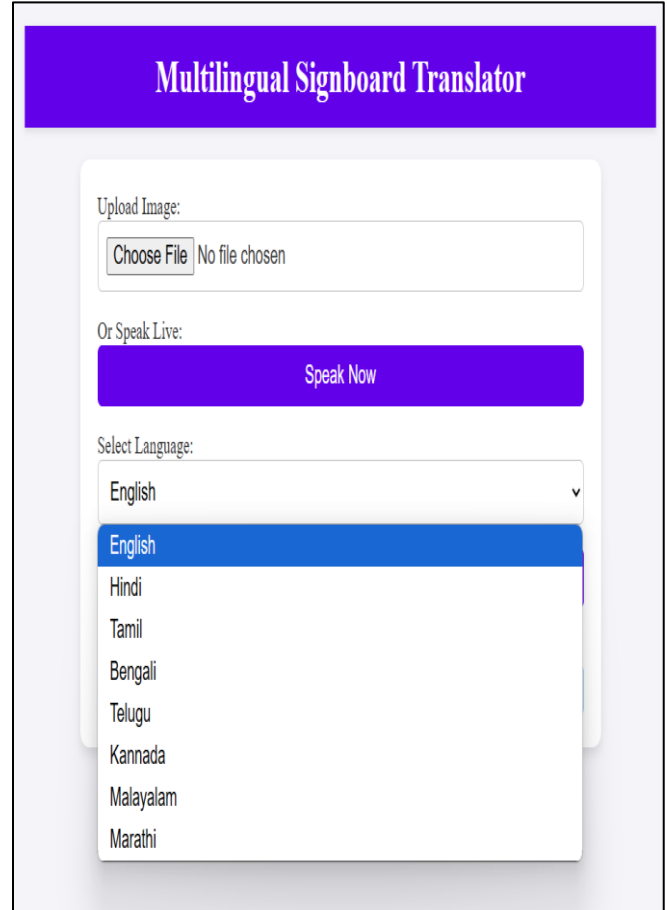


Fig. 2: The web interface

Fig. 2 showcases the web interface of the multilingual signboard translator. It presents a simple, user-friendly design where the user can easily interact with the system. The interface includes a file upload option for images, a button to input live audio, and a dropdown menu for selecting the target translation language. In this image, the dropdown menu is visible, showing a list of eight supported languages: English, Hindi, Tamil, Bengali, Telugu, Kannada, Malayalam, and Marathi. This multi-language support enables the system to cater to a wide range of regional signboards, making it suitable for diverse linguistic contexts. The interface layout is clean and intuitive, ensuring a seamless user experience.



Fig. 3: Input 1

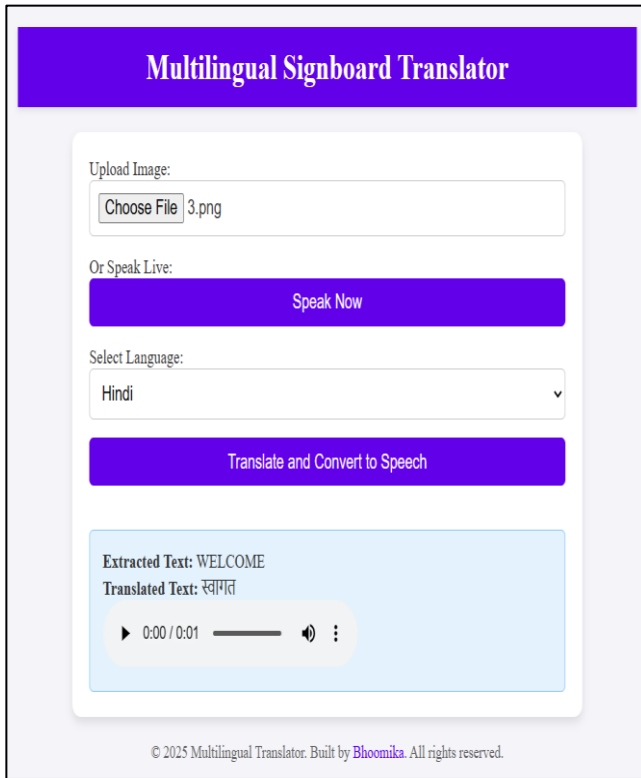


Fig. 4: Output 1

Fig. 4 displays the result after an image with a “WELCOME” signboard is uploaded into the system. The uploaded image contains text in English, which is extracted by the system using OCR. The output is then translated into Hindi, as selected by the user from the language dropdown. The translated text is displayed as “स्वागत”, showcasing the system’s ability to process and accurately translate the extracted text into the target language. It also provides the audio output in Hindi, demonstrating how the system handles images and provides accurate multilingual translations in real-time.



Fig. 5: Input 2

In Fig. 6, a signboard in Hindi “मुंबई” is uploaded to the system. The OCR extracts the Hindi text, and then the system translates it into Tamil, as “மும்பை”. The Tamil translation is displayed in the output section of the interface, and the speech is also provided in Tamil. This image highlights the system’s multilingual capabilities, as it seamlessly converts text from one Indian language to another. It also demonstrates how the system can handle regional languages and offer accurate translations, which is particularly useful for users who need translations for signage in different languages, such as in the case of a diverse linguistic landscape.

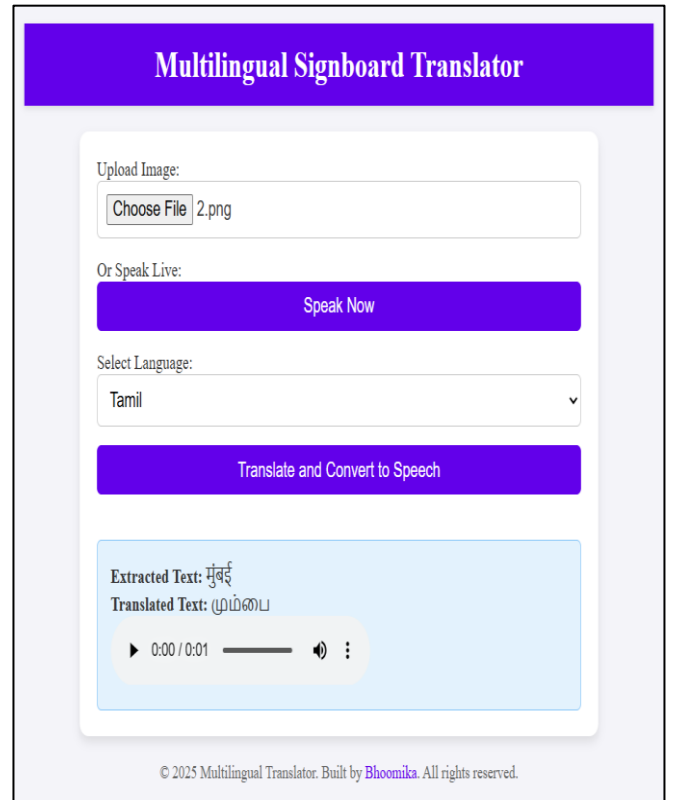


Fig. 6: Output 2

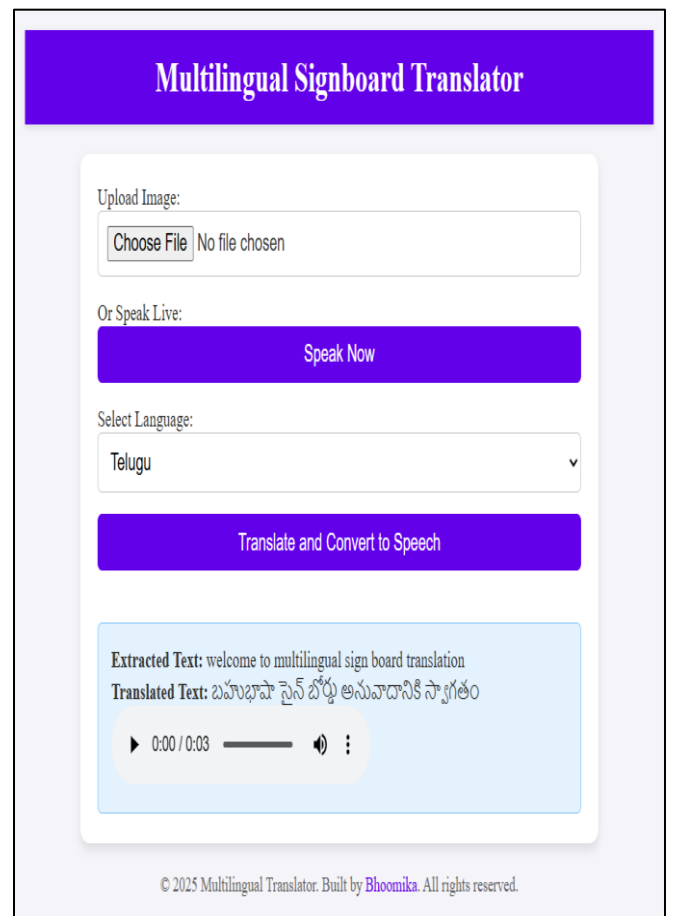


Fig. 7: Output 3

Fig. 7 captures the scenario where the user provides a voice input instead of an image. The voice input, “welcome to multilingual signboard translation”, is spoken by the user,

and the system transcribes the speech using Speech Recognition. After transcribing the input, the system translates the text into Telugu, as selected by the user. The translated text, “బహుభాషా సైన్ బోర్డ్ అనువాదానికి స్వాగతం”

is then displayed on the interface. In addition to the translated text, the system also generates speech output, allowing the user to hear the translated message in Telugu. This image highlights the system’s versatility in handling voice input and providing real-time speech-to-text conversion, followed by accurate language translation.

VII. CONCLUSION

The multilingual signboard translation system successfully combines Optical Character Recognition (OCR), speech recognition, and language translation to provide an efficient and user-friendly tool for real-time multilingual translation. By offering both image-based and voice input options, the system can cater to various user needs, ensuring that it can extract and translate text from signboards in different languages accurately. The integration of Indian languages further enhances their practical use, especially in diverse multilingual regions, allowing for seamless communication and understanding of local signage.

This system demonstrates the powerful capabilities of modern technologies in bridging language barriers and providing instant translations. It not only enhances the accessibility of regional signboards for people from different linguistic backgrounds but also offers a versatile solution for travelers and residents alike. Ultimately, the system showcases the potential of combining OCR, speech recognition, and machine translation to create a practical, real-world application for language translation.

VIII. FUTURE SCOPE

The future scope of this multilingual signboard translation system includes expanding support for more regional and international languages, improving OCR accuracy for complex or stylized fonts, and enhancing speech recognition to handle diverse accents and noisy environments. Integration with mobile platforms and real-time camera translation features can further increase usability for on-the-go users. Additionally, incorporating AI-driven context detection and offline translation capabilities would make the system more robust and reliable, especially in remote or low-connectivity areas. Also, including gesture-based controls along with personalized voice styles and accents, and adding language learning features for more efficiency.

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